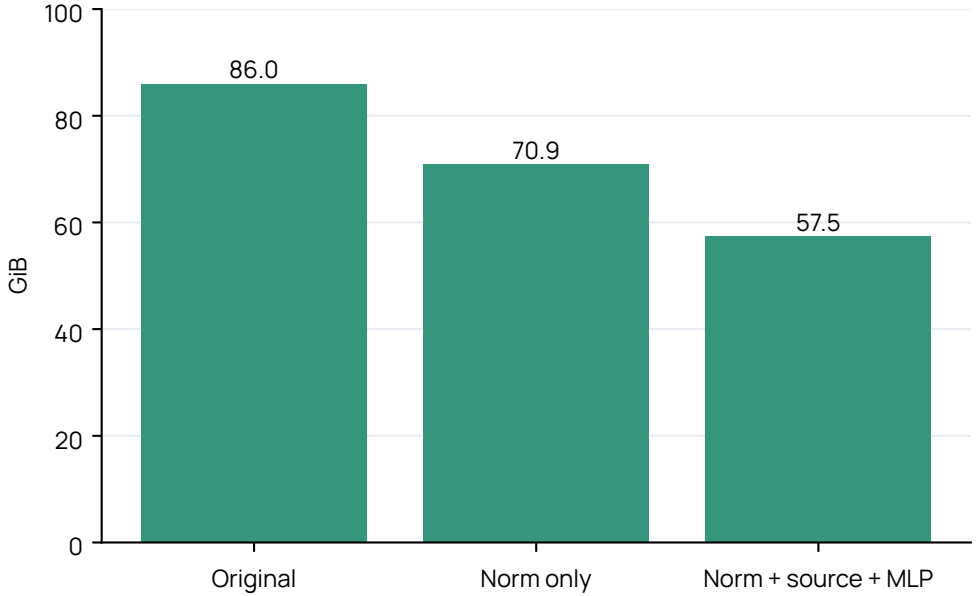
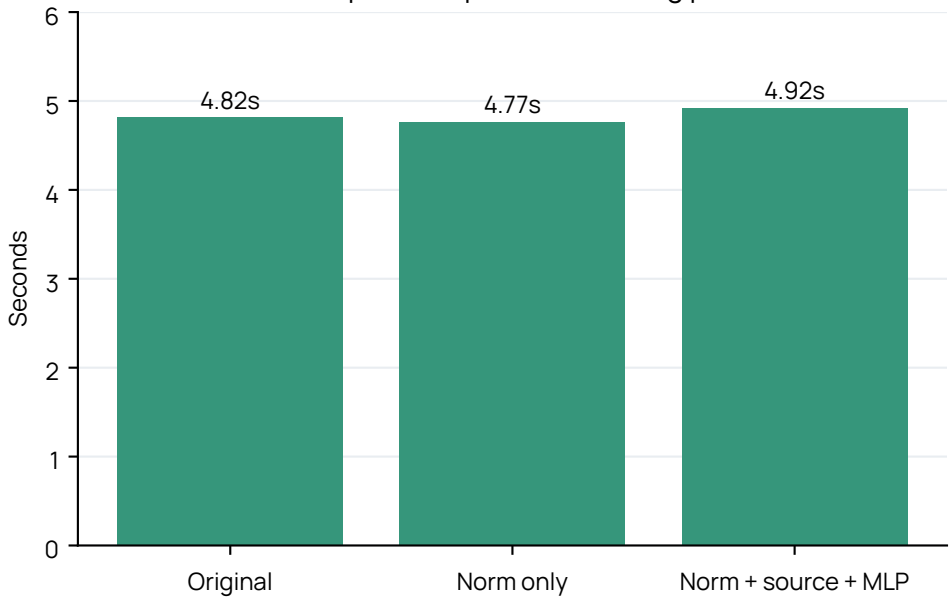
Lossless chunking diagnostics · exact checkpoint · batch 128 · microbatch 32

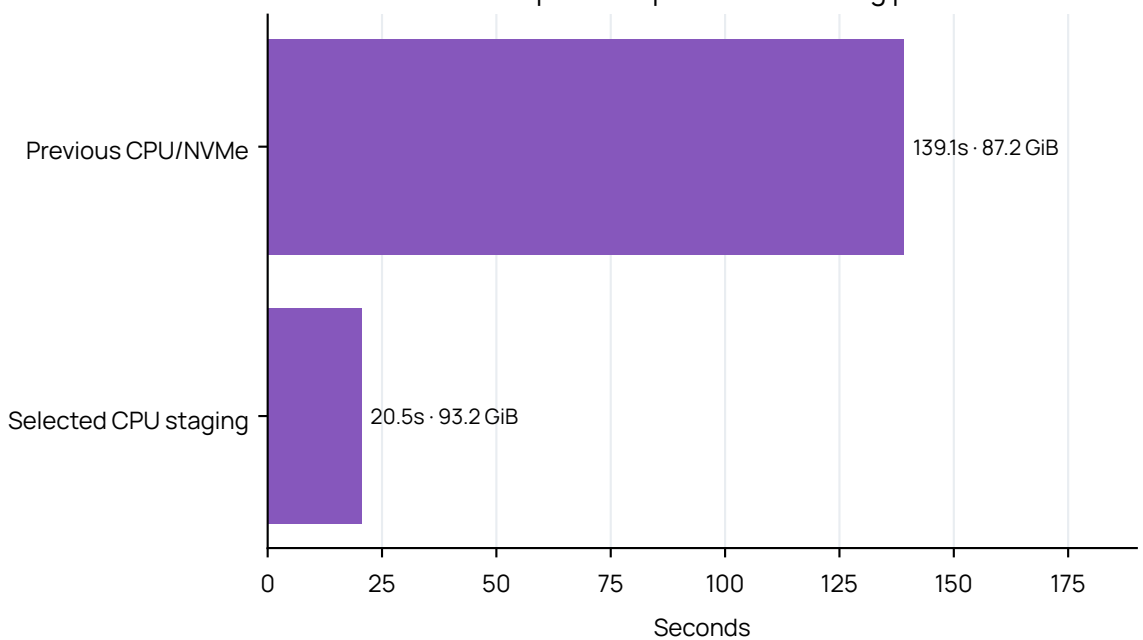
43k complete step · allocated GPU memory



43k complete step · timed training phases



128k complete steps · timed training phases



Complete outputs and all 349 parameter gradients

Variant	Output exact	Gradient ΔL2
Original repeated	True	1.4e-08
Norm only	True	1.4e-08
Norm + source + MLP	True	1.3e-08
128k validated	True	8.2e-09

128k reference: original operators with checkpoint/offload wrapper.
Broad whole-entry chunking rejected for numerical differences.
Total memory still scales with vocabulary size.